flop7

The unofficial Flip 7 terminal emulator

> NOTE: This document was created with the help of Github Copilot. The architecture itself was created by Ben Halladay, and Copilot assisted in generating diagrams based on code already written by Ben Halladay.

Overview

flop7 is a terminal based emulator for the popular game Flip 7. It allows the user to track points for a live game, or play the game virtually against others or against bots.

Flip 7 — Game Summary

Flip 7 is a press-your-luck card game for 3+ players. Players race to be the first to accumulate **200 points** across multiple rounds.

The 94-card deck contains three categories of cards:

Category	Cards	Bustable
Number (0–12)	79 cards — each value N appears N times (except 0, which has 1)	Yes
Action (Flip Three, Freeze, Second Chance)	9 cards (3 of each)	No
Score Modifier (+2, +4, +6, +8, +10, ×2)	6 cards (1 of each)	No

Each round, players take turns choosing to **hit** (draw a card) or **stay** (bank their points and exit the round). Drawing a duplicate number card causes a **bust** — the player scores 0 for that round. If a player collects 7 unique number cards, the round ends immediately and they receive a 15-point bonus.

Action cards allow you to target other players with special abilities: **Flip Three** forces a target to draw 3 cards, **Freeze** forces a target to bank and exit, and **Second Chance** absorbs one future duplicate.

Score modifier cards adjust end-of-round scoring: ×2 doubles the number card total (applied first), then flat bonuses (+2 through +10) are added on top.

Repo structure overview

The codebase is organized into four domain layers plus entry points:

```
src/flop7/
  __main__.py              # Entry point for python -m flop7
  cli.py                   # Entry point for flop7 command-line tool
  app/                     # Top-level orchestration layer
  tui/                     # Terminal user interface (urwid-based)
  bot/                     # Bot models and decision-making strategies
  core/                    # Game rules engine and data models
```

Layer responsibilities

- **`app/`** — Orchestrates the application lifecycle and routes user commands to game logic
- **`tui/`** — Handles all terminal rendering and user input via urwid
- **`bot/`** — Provides bot player models with configurable decision strategies
- **`core/`** — Implements the game engine, card definitions, and rules
- **`cli.py`** / **`__main__.py`** — Bootstrap the application

App design

The `app/` layer is the top-level orchestration layer. It is responsible for starting the TUI loop, receiving user commands, and routing those commands into game setup / gameplay state transitions.

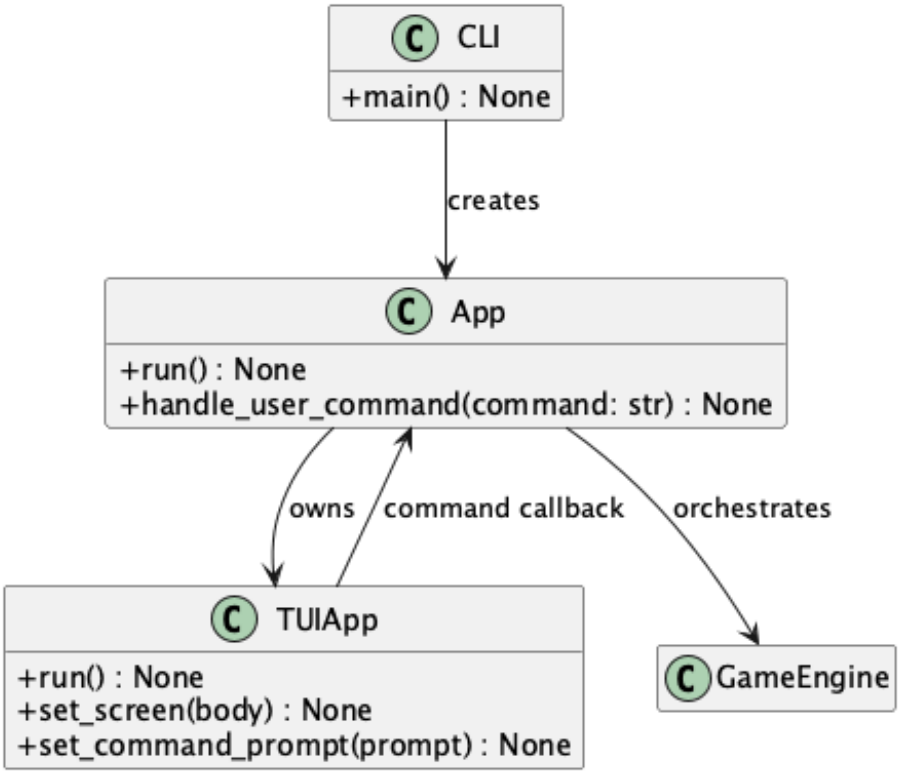
Responsibilities

1. Construct the TUI and provide a command handler callback
2. Start and stop the application lifecycle (`run`, exit behavior)
3. Coordinate screen/state transitions at the application level
4. Act as the bridge between command input and game logic entry points

Control flow

1. `cli.py` creates `App` and calls `run()`.
2. `App` constructs `TUIApp`, injecting `handle_user_command` as the callback.
3. `TUIApp` runs the urwid `MainLoop`.
4. On Enter, the command bar text is sent back to `App.handle_user_command(...)`.
5. `App` interprets the command and triggers app-level behavior (navigation, setup, game actions).

App orchestration diagram



Core design

Architecture

The game logic lives in `src/flop7/core/` and is organized into four sub-packages:

```
core/
  classes/      # Data objects - Card, Deck, Player
  engine/       # Game loop and action card resolution
  enum/         # Enumerations for events and decisions
  protocols/    # Structural typing contracts (Protocol classes)
```

`classes/` — Data Layer

- **`Card`** — A `@dataclass` representing a single card definition. Fields include the card's name, abbreviation, deck count, point value, bustable flag, an optional `special_action` callback, and an optional `score_modifier` callback with an associated `score_priority`. All 22 unique card types are

defined as module-level constants in `cards.py` (e.g. `ZERO`, `FLIP_THREE`, `PLUS_TWO`), and collected into `ALL_CARDS` and `CARD_MAP` for lookup.

- **`Deck`** — Manages the draw pile and discard pile. On construction it expands `ALL_CARDS` into the full 94-card list (respecting `num_in_deck`) and shuffles. Drawing is delegated to an injected `DrawProtocol` callable — this allows the same `Deck` class to serve both a virtual game (random pop) and a physical-card mode (external input). `reshuffle()` recycles the discard pile when the draw pile runs low.
- **`Player`** — Pure state container for a single player. Tracks `name`, `hand`, cumulative `score`, and `is_active` status. The computed property `active_score` calculates the current round score by sorting modifiers by `score_priority` ($\times 2$ first, then flat bonuses) and folding them over the number-card sum.

`engine/` — Game Loop

- **`GameEngine`** — The core engine class that owns the round lifecycle. It holds the `Deck`, player list, and two injected decision callbacks (`HitStay` and `TargetSelector`). The core loop is:
  1. `play()` calls `round()` until a player reaches 200 points.
  2. `round()` iterates active players: each either hits or stays.
  3. `hit()` processes a drawn card — checking for Second Chance absorption in `pre_hit_hook()`, dispatching `special_action` callbacks for action cards, and detecting busts for duplicate number cards.
  4. After the round, every player's `active_score` is added to their cumulative `score`, hands are discarded, and the win condition is checked.
- **`actions`** — Module-level functions (`flip_three`, `freeze`, `second_chance`) that implement the three action card effects. Each follows the `CardAction` protocol signature `(game, player, card) -> bool`. At module load time, these functions are patched onto the corresponding `Card` constants via `FLIP_THREE.special_action = flip_three`, etc.

`enum/` — Enumerations

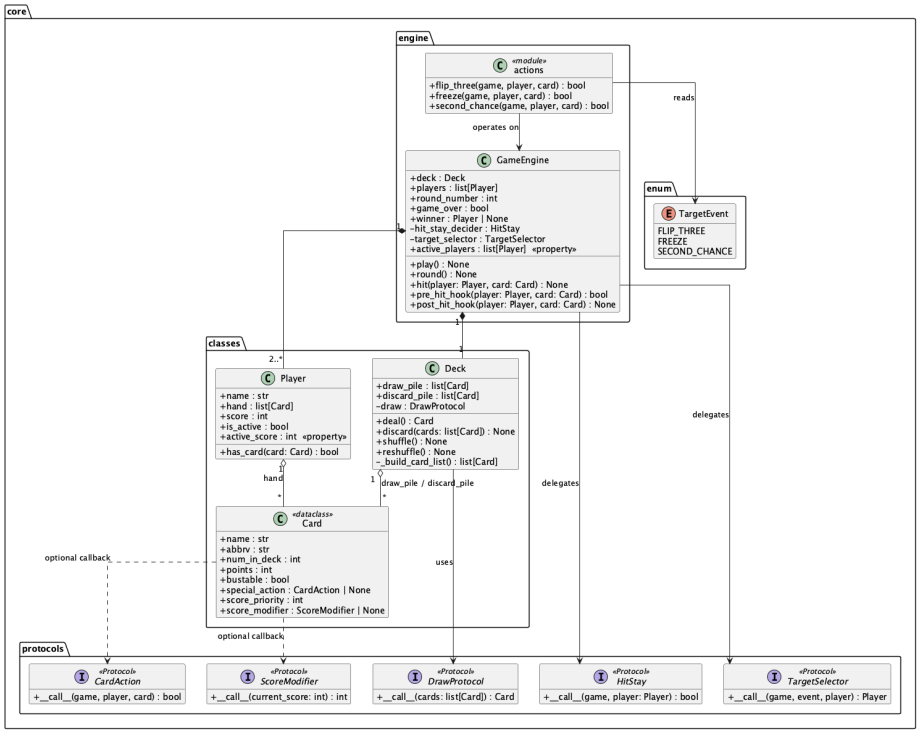
- **`TargetEvent`** — Identifies the reason a target is being selected (Flip Three, Freeze, or Second Chance), so the `TargetSelector` callback can present appropriate context.

`protocols/` — Dependency Contracts All external behavior the engine depends on is expressed as `typing.Protocol` classes, enabling structural (duck) typing:

Protocol	Signature	Purpose
`DrawProtocol`	`(cards: list[Card]) -> Card`	How the deck draws a card (random vs. user-input)
`HitStay`	`(game, player: Player) -> bool`	Decides whether a player hits or stays with access to full game context
`TargetSelector`	`(game, event, player) -> Player`	Picks a target for action cards by deriving candidates from game state
`CardAction`	`(game, player, card) -> bool`	Executes a card's special effect, tied to the `Card` class
`ScoreModifier`	`(current_score: int) -> int`	Transforms score (e.g. $\times 2$, $+4$), tied to the `Card` class

This protocol-based design decouples the engine from any concrete UI or strategy implementation — the same `GameEngine` can be driven by a TUI, a bot, or automated tests simply by injecting different callables.

Class Diagram



Interface

The terminal interface is intentionally simple: one main content area and one persistent command area at the bottom. The command area is the same on every screen, so users always interact with the app in a consistent way.

Visual mockups

Home screen mockup:

Game screen mockup:

Layout model

The UI is split into two vertical zones:

1. **Main view (top):** screen-specific content
2. **Command console (bottom):** output + input + suggestions

The main view changes based on context (home vs game), but the command console stays persistent.

Figure 1: Home screen mockup

Figure 2: Game screen mockup

Home screen (simplified)

The home screen is mostly a landing page:

- Centered ASCII title/logo for visual identity
- Short subtitle/tagline beneath the logo
- No complicated widgets in the body
- All interaction happens through the command console

Expected commands from home are simple navigation commands (for example: start real game, start virtual game, help, quit).

Game screen (simplified)

The game screen keeps the same console at the bottom, but the top area becomes a live game board:

- Player panels are shown in a grid-like row/stack layout
- Each panel shows the player's visible cards and name
- The active player's panel can be visually emphasized
- Score and turn context are shown in the main area, not inside the input line

This gives players one stable mental model: watch the board above, type commands below.

Command console behavior

The command console has three parts (top to bottom):

1. **Output line/block** for short status text and prompts
2. **Input line** (prefixed with `>`) where the user types commands
3. **Suggestion list** with up to three visible completions

If more than three completions are available, the user navigates them with arrow keys instead of expanding the list indefinitely. The main view above should resize as needed to preserve console visibility.

Interaction philosophy

The TUI is command-driven rather than mouse-driven:

- The user always types explicit commands
- The system responds with clear textual prompts
- Screen changes are lightweight and immediate

This design keeps the interface fast, accessible over any terminal, and easy to test because the input/output contract is text-first.

Virtual Bots

The bot system is designed as a strategy layer that plugs directly into the core engine decision protocols.

Design goals

1. Support multiple bot models with minimal boilerplate
2. Keep bot decision logic isolated from TUI and orchestration
3. Allow advanced models (like omniscient) in virtual mode only
4. Make model registration and selection simple for setup screens

Planned architecture

Bot logic lives in `src/flop7/bot/`.

```
bot/
  base.py                  # AbstractBot contract
  registry.py              # model-name -> bot-class lookup
  knowledge.py             # shared game-state views (planned)
  utils.py                 # shared probability helpers (planned)
  models/
    basic.py             # baseline probability-driven bot
    omniscient.py        # full-information virtual-only bot
```

Base bot contract

All bot models implement `AbstractBot` in `bot/base.py`:

- `hit_stay(game, player) -> bool`
- `target_selector(game, event, player) -> Player`

These signatures match the core decision protocol style (the full game object is passed in), so each model can inspect round state, scores, and active players before deciding.

Model registry

`bot/registry.py` provides the `Bot` class, a central registry and factory for bot model instantiation. It maintains a class-level mapping of model names to implementations:

- `Basic` -> `BasicBot`
- `Omniscient` -> `OmniscientBot`

Instantiation is handled via the constructor:

```
Bot(model="Basic", virtual=False, **params)
```

The `virtual` flag is used to validate that virtual-only models are not used in real (non-virtual) games. This keeps setup/UI code decoupled from model class

imports — the interface layer can request a bot by name without knowing the concrete class.

Model behavior (initial set)

Basic bot The baseline model targets straightforward, explainable strategy:

- Estimate bust risk before hitting
- Stay when bust risk exceeds a threshold
- For action cards, target the strongest active opponent (highest score / board pressure)

This model is intended to be predictable and easy to benchmark.

Omniscient bot The omniscient model is a benchmarking model for virtual play:

- Reads full hidden deck state
- Computes exact or near-exact bust probabilities from remaining cards
- Uses deeper expected-value decisions than Basic

Because it uses hidden information, it should be restricted to virtual games and never used for live/real deck tracking mode.

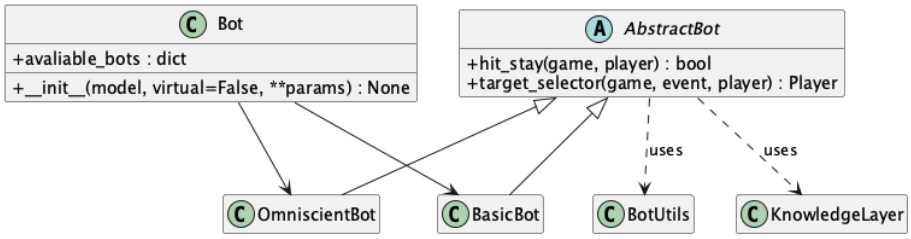
Knowledge and utility layers (planned)

Two shared modules are scaffolded for modularity:

- `knowledge.py`: builds reusable game-state views (public-only vs omniscient)
- `utils.py`: common helper math (bust probability, score pressure heuristics, tie-break logic)

The intent is to prevent duplicate math and state-derivation code across models.

Bot UML diagram



Sequence overview

1. Game engine requests a `HitStay` or `TargetSelector` decision.
2. The configured bot model is called via the `AbstractBot` contract.

3. The model evaluates game state (and, where relevant, helper layers) to choose an action.
4. The selected action is returned to the engine through the protocol callback path.